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Emotion recognition from spatiotemporal EEG representations with hybrid convolutional recurrent neural networks via wearable multi-channel headset

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ABSTRACT

Emotion recognition from electroencephalography (EEG) has been an important research direction in affective brain–computer interactions (ABCI). An integration of EEG-based emotion recognition algorithms in ABCI can make the users experience more complete and engaging. In this paper, we propose a new data representation of electroencephalogram (EEG), which transforms 1D chain-like EEG vector sequences into 2D mesh-like matrix sequences. The mesh structure of the matrix at each time point corresponds to the electrodes' location of EEG headset, which could better represent the spatial correlation of EEG signals among multiple physically adjacent electrodes. Then, the sliding window is used to divide the 2D matrix sequences into segments containing equal time points, and each segment is seen as an EEG sample integrating the temporal and spatial correlation of raw EEG recordings. We also propose both cascaded and parallel hybrid convolution recurrent neural networks to accurately predict the emotional category of each EEG sample. Extensive binary emotion classification experiments in valence and arousal are carried out on a large scale DEAP dataset (32 subjects, 9,830,400 EEG recordings). The experimental results demonstrate that the classification accuracies of both proposed hybrid networks on our spatial–temporal EEG representation achieve over 93%, which outperform the most recent baseline methods and other deep learning models in within-subject validation scenario. Our proposed methods and hybrid models effectively improve the accuracy and robustness of EEG emotion classification, which can be applied to develop affective BCI applications using consumer multi-electrode EEG headset.

1. Introduction

In recent years, with the development of deep learning and artificial intelligence technology, emotion recognition has become a hot research topic in the field of human–computer interaction and affective computing. Emotion recognition is the process of understanding the emotion type expressed by a person. The development of efficient and robust human emotion recognition algorithms will have a significant impact on wearable human–computer interaction [1]. The introduction of automatic emotion recognition into human–computer interaction applications can significantly improve the quality of user experience, bring more wearable equipment for emotional EEG sensing, and realize many control functions based on emotion perception and regulation. Electroencephalogram (EEG) is a non-invasive brain imaging technology, which relies on the scalp electrodes to measure the electrophysiological activities of the brain, and contains a lot of physiological, psychological and pathological information. A large number of neurophysiological and psychological studies have found that the generation and activity of human emotions are highly correlated to the activity of cerebral cortex. Different cognitive and emotional activities of human beings can induce different EEG signals. Effective feature extraction and classification can help achieve the purpose of affective brain–computer interface (BCI) control [2]. Compared with human facial expression, voice, electrocardiogram (ECG), electromyogram (EMG), magnetic resonance imaging (MRI) and other physiological signals, EEG is becoming more popular for human emotion recognition because of its advantages of strong objectivity, not easy to forge, easy to acquire with wearable EEG headset and easy to operate [3].

The processing and recognition of EEG signal is a great challenging task. Firstly, the EEG signal is of low signal-to-noise ratio, and it is fragile to be interfered by any other noise. For example, sensitive EEG recording equipment is very easy to be influenced by the surrounding environment and muscle activity, eye movement or blinking may also bring undesired noise. Secondly, people are often interested in EEG signals relating specific brain activities which always submerged in background noise, and it is difficult to separate these signals from the background noise. Although the spatial resolution of EEG signal on the scalp is low, its high time resolution (in milliseconds) can record the slow or rapid changes of brain activity [4]. Therefore, to extract

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features related to specific brain emotion dynamics, both spatial and temporal EEG correlation need to be considered.

In recent years, deep learning technology has achieved great success in recognition tasks in computer vision, automatic speech recognition, natural language processing, bioinformatics and other fields [5–7], due to its strong ability of end-to-end self-learning of complex high-level feature representation, thus avoids much manual feature extraction work. Convolutional neural networks (CNNs) is the core of the current best architecture for recognizing image and video data, because they can learn feature representations that are robust to partial translation, rotation, scaling and deformation of input data [8]. On the other hand, long-short term memory (LSTM) based recurrent neural networks (RNNs) have shown the most advanced performance in recognizing time series dynamics, such as text and speech recognition [6]. In addition, Ng. et al. [9] combined these two types of networks for video classification and achieved good performance.

With the emergence of portable EEG acquisition technology and equipment, large-scale EEG data becomes easier to obtain, and deep learning technologies such as CNN, RNN and LSTM are increasingly applied to the classification of EEG time series. Bashivan et al. [10] proposed a deep recursive convolutional neural network (R-CNN) for EEG-based cognitive and mental load classification task. With R-CNN, they captured robust features that were invariable in time, space and frequency, and their classification error rate was reduced by more than 50% compared with their previous support vector machine (SVM) and random forest models. Ryan et al. [11] used a LSTM based deep RNN model to explain the time dependence of cognitive-related EEG signals, which significantly improved the stability of cross-day EEG features and achieved an average classification accuracy of 93% in within-subject cognitive status recognition scenario. Dalin et al. [12] proposed a deep convolution recurrent neural network model to learn the spatiotemporal robust feature representation of the raw EEG data stream, and achieved the accuracy of 98.3% for within-subject motion intention classification in large scale MI-EEG dataset, which outperformed other simple classifiers and deep learning models. Lawhern et al. [13] proposed an application of multi-layer pure convolution neural network without full connection layer and achieved the best performance in P300-based oddball recognition task, motor-related cortical potential recognition in finger motor task and sensory motor rhythm recognition in motor imagination task. All the above methods provide valuable references and inspirations for EEG-based emotion recognition.

Recently, Alhagry et al. [14] proposed a deep RNN model based on LSTM structure and got the average emotion classification accuracy of 85.65%, 85.45% and 87.99% in dimensions of arousal, valence and liking respectively on large scale DEAP dataset, which outperformed traditional methods. Soleymani et al. [15] applied LSTM-RNN model and continuous conditional random field (CCRF) algorithm to detect the real emotional state of the subjects from their EEG signals and facial expressions when they were watching videos and achieved better results. Salama [16] et al. used a 3D-CNN to recognize human emotions from multichannel EEG data of DEAP dataset, and obtained the recognition accuracy of 87.44% and 88.49% in valence and arousal respectively. In our previous study [17], we also applied deep CNN model to extract the invariable high-level representations of the time, frequency and their combined features of the EEG signals in DEAP dataset, and achieved the binary recognition accuracy of 88.5% and 86.7% in valence and arousal respectively. In [18], we proposed a hierarchical bidirectional Gated Recurrent Unit (GRU) network with attention for human emotion classification from continues EEG signals of DEAP dataset, which effectively reduced the impact of long-term non-stationarity of EEG sequences. All above methods have improved the performance of EEG emotion classification to a certain extent, but there are few models that can jointly learn the discriminative EEG features correlated to human emotions from spatial, temporal and frequency domain. Therefore, the accuracy and robustness of EEG emotion recognition is still need to be further improved.

This paper proposes a new method of EEG data representation, which converts 1D chain-like EEG vector sequences into 2D mesh-like matrix sequences. The matrix structure just maps the brain area distribution of 32 EEG electrodes' location which could better represent the spatial correlation of multiple physically neighboring electrodes. Then, the sliding window is used to divide the 2D matrix sequences into segments including equal time points. Each segment is called an EEG sample or epoch integrating full spatial and temporal information. Both cascaded and parallel convolution recurrent neural networks are proposed to predict the emotion category of each EEG sample. In these two hybrid networks, CNN is used to learn the high-level spatial correlation between physically adjacent EEG electrodes, and LSTM based RNN is used to learn the subtle temporal dependency between time points. Extensive binary emotion classification experiments in valence and arousal are carried out on a large scale DEAP dataset (32 subjects, 9,830,400 EEG recordings) to verify the effectiveness of our proposed methods and models. The experimental results demonstrate that the classification accuracies of both proposed hybrid networks on our spatial–temporal EEG representations achieve over 93%, which outperform the most recent baseline methods and other deep learning models and yield accuracy increase of 5.1% and 6.6% separately in valence and arousal in within-subject validation scenario.

2. Methods

In this section, we describe the method of converting 1D chain-like EEG sequences to 2D mesh-like matrix sequences and explain the detailed architectures and working processes of our proposed cascaded and parallel convolutional recurrent network models.

2.1. 2D mesh-like EEG sequences transformation

Most EEG-based affective BCI systems usually use portable wearable multi-electrode headset to collect EEG signals. When the subject wearing this equipment watches the stimulus video, the sensing electrodes on the headset capture the current fluctuation of the brain scalp. The whole process of EEG signals acquisition and proposed EEG data transformation method is shown in Fig. 1, in which the EEG electrode map shows the electrode position distribution of the headset used in DEAP experiment. The number and distribution of electrodes of EEG headset in different BCI systems are different. The readings obtained by the EEG acquisition headset represent the time series of EEG potentials at a certain sampling frequency. Usually, the raw EEG signal collected at time point t is represented with a 1D chain-like vector as $X_t = [x_t^1, x_t^2, \ldots, x_t^n]^T$, where n is the number of electrodes or channels of the EEG headset, and x_t^n is the reading of the nth electrode at time point$t$. For the observation period $[t, t+N-1]$, there are total N such data vectors each of which contains n elements corresponding to the reading of n electrodes.

EEG signals are collected using a multiple-electrode headset and recorded as 1D chain-like vector series. These vectors are then transformed into 2D mesh-like matrix series according to the electrode map of the EEG acquisition headset. The converted mesh series are then segmented into clips using sliding window. From the EEG electrode map in Fig. 1, it can be seen that each electrode is physically adjacent to multiple neighboring ones which are often used to measure the EEG dynamics of certain brain area. Different brain areas correspond to different brain activities. The raw 1D chain-like EEG vector can only represent the correlation between two adjacent electrodes. Therefore, we try to convert the original 1D chain-like EEG vector sequences into 2D mesh-like matrix sequences according to electrode distribution of EEG acquisition headset. In this paper, the 32-channel EEG vector X_t

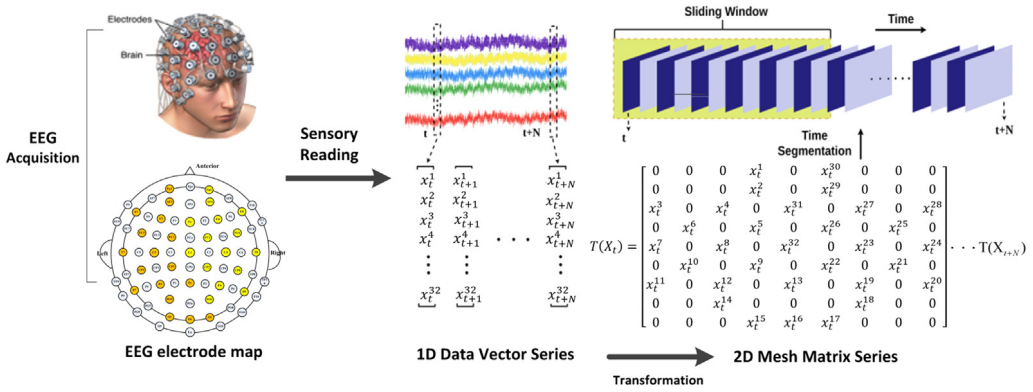


Fig. 1. EEG signal acquisition and transformation process.

is transformed into a 2D mesh matrix Y_t as follows:

$$Y_t = T\left(X_t\right) = \begin{bmatrix} 0 & 0 & 0 & x_t^1 & 0 & x_t^{30} & 0 & 0 & 0 \\ 0 & 0 & 0 & x_t^2 & 0 & x_t^{29} & 0 & 0 & 0 \\ x_t^3 & 0 & x_t^4 & 0 & x_t^{31} & 0 & x_t^{27} & 0 & x_t^{28} \\ 0 & x_t^6 & 0 & x_t^5 & 0 & x_t^{26} & 0 & x_t^{25} & 0 \\ x_t^7 & 0 & x_t^8 & 0 & x_t^{32} & 0 & x_t^{23} & 0 & x_t^{24} \\ 0 & x_t^{10} & 0 & x_t^9 & 0 & x_t^{22} & 0 & x_t^{21} & 0 \\ x_t^{11} & 0 & x_t^{12} & 0 & x_t^{13} & 0 & x_t^{19} & 0 & x_t^{20} \\ 0 & 0 & x_t^{14} & 0 & 0 & 0 & x_t^{18} & 0 & 0 \\ 0 & 0 & 0 & x_t^{15} & x_t^{16} & x_t^{17} & 0 & 0 & 0 \end{bmatrix} \quad (1)$$

Where, t is a specific time point and the unused electrode position is set to 0 which does not work in neural networks. Through conversion, the 1D chain-like EEG vector sequence $[X_t, X_{t+1}, \ldots, X_{t+N-1}]$ in observed period $[t, t+N-1]$ is transformed into a 2D matrix sequence $[Y_t, Y_{t+1}, \ldots, Y_{t+N-1}]$, where the number of meshes is still N. The non-zero data in each matrix is then normalized by Z-score algorithm. Next, as shown in the last step in Fig. 1, a sliding window is used to segment the 2D matrix sequence into multiple individual epochs containing not only temporal information, but also more spatial information related to the brain emotional activity at each time point. Each EEG epoch is represented as $P_j = [Y_t, Y_{t+1}, \ldots, Y_{t+s-1}]$, which has the same length as size of the sliding window and there is no overlap between the two adjacent clips, where $j = 1, 2, \ldots, q$, q is the number of segmented EEG epochs in an observation period and$s$ represents the size of the sliding window, also the number of time points in each epoch. In the followings, we will further propose effective hybrid deep learning models to learn the high-level discriminative spatiotemporal features from 2D matrix sequences of each EEG epoch P_j and predict the corresponding emotional category of that.

2.2. Cascaded hybrid convolutional recurrent neural network

We first propose a hybrid EEG emotion classification model based on a cascaded convolution recurrent neural network (CASC-CNN-LSTM for short), which architecture is shown in Fig. 2. The input of the model is a 2D mesh-like EEG matrix sequence, such as an EEG sample P_j. A 2D CNN network is firstly applied to extract the high-level spatial feature representation from the 2D mesh-like matrix at each time point. The extracted spatial feature representations are then input into a LSTM-based RNN network to further extract the temporal features. The output of the LSTM network is fed into a full connection layer and its output is finally fed into a SoftMax layer for the final emotion prediction.

As shown in Fig. 2, we first use a 2D CNN network to learn the high-level spatial feature representation from the 2D mesh-like matrix at each time point. The jth input EEG epoch is expressed by a 2D

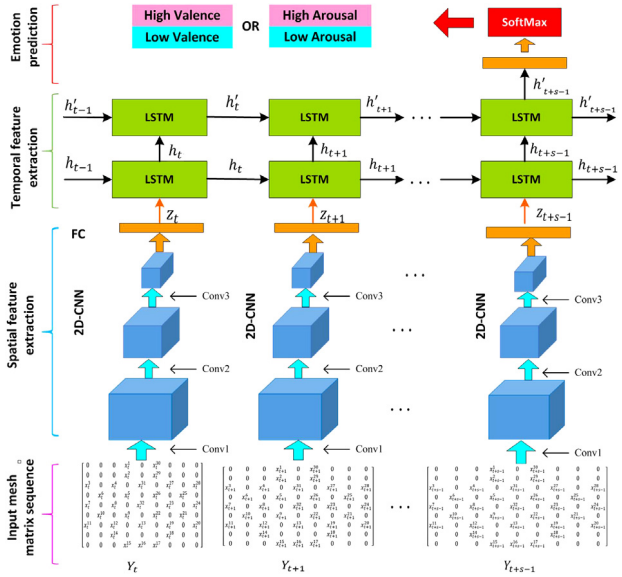


Fig. 2. The architecture of cascaded hybrid convolutional recurrent neural network.

matrix sequence $P_j = [Y_t, Y_{t+1}, \ldots, Y_{t+s-1}] \epsilon R^{s \times h \times w}$, which includes s elements and each element $Y_k\,(k = t, t+1, \ldots t+s-1)$ is a 2D mesh-like matrix whose size is $h \times w$, where h, w represents the height and width of the mesh matrix respectively. The mesh matrix at each time point is input into a 2D CNN network, and the learned spatial feature representation is expressed as: $Z_k\,(k = t, t+1, \ldots t+s-1)$: $Z_k = CNN_{2D}\left(Y_k\right)$, $Z_k \epsilon R^l$. Z_k is a feature vector containing l elements. Thus, the input EEG matrix sequence is transformed into the sequence of the spatial eigenvector: Casc-2DCNN: $P_j \rightarrow Q_j$, where $Q_j = \left[Z_t, Z_{t+1} \ldots, Z_{t+s-1}\right] \in R^{s \times l}$. Our proposed 2D CNN model consists of three convolution layers, each layer has 32, 64 and 128 convolution kernels all with size of 3 × 3 to perform non-padding convolution operation. ReLu activation function is used at each layer, the learning rate is 0.002 and Adam optimizer is used. The first convolution layer has 32 feature maps with size of 7 × 7, the second convolution layer has 64 feature maps with size of 5 × 5, and the last convolution layer gets 128 feature maps with size of 3 × 3 which are then input into a full connection layer containing 1024 neurons to transform these 128 feature maps into the final spatial characteristic representation $Z_k \epsilon R^{1024}\,(k = t, t+1, \ldots t+s-1)$. The full connection layer between 2D CNN and LSTM is optional, but the experimental results show that adding the full connection layer plays an important role in helping the model convergence and improving the performance of the whole framework.

Next, we use LSTM to build a two-layer RNN model which accepts the spatial feature representation sequence Q_j as input to further extract the temporal correlation features of each EEG epoch. LSTM is an improved neuron structure of RNN hidden layer to solve the problem of gradient disappearance and gradient explosion. Each layer of the model contains s LSTM units. The hidden state of the LSTM unit in the first layer at the current time point t is represented as h_t, then h_{t-1} represents its previous hidden state at time point *t-1*. In each layer, the information of the previous time point is transferred to the current time point, and so on, affecting the final output at the last time point. The hidden state sequence $[h_t, h_{t+1}, \dots h_{t+s-1}]$ output by the first RNN layer is input into the corresponding LSTM cells in the second layer. Because we are interested in the emotional category of the whole EEG epoch, the output h'_{t+s-1} of the last time point in the second layer extracted after learning all the time points in the full window is taken as the temporal feature learned by the whole RNN network, which is sent to the next full connection layer, as shown in the last stage of Fig. 2. This extracted temporal feature h'_{t+s-1} is represented as: Cas-RNN: $h'_{t+s-1} = RNN_{lstm}(Q_j)$, $h'_{t+s-1} \in R^i$, where i is the size of the hidden state of a LSTM cell. Finally, after the full connection layer, a SoftMax layer is appended to predict the probability of each emotion category: SoftMax: $C_j = SoftMax\left(h'_{t+s-1}\right)$, $C_j \in R^k$, where k is the number of emotion categories to be predicted. After the full connection layer in the classification stage, we also use dropout operation to make feature regularization.

In general, our cascaded hybrid framework sequentially transforms the raw 1D chain-like EEG vector stream in the observation period into 2D mesh-like matrix sequence, segments them into multiple epochs of equal length, extracts the spatiotemporal high-level features by CNN and LSTM networks, and at last classifies each epoch into different emotion categories. Each EEG epoch P_j contains s transformed 2D matrices $[Y_t, Y_{t+1}, \dots, Y_{t+s-1}]$. A 3-layer 2D CNN network is used to learn the spatial feature representation from each mesh matrix. From the learned spatial feature sequence $[Z_t, Z_{t+1}, \dots, Z_{t+s-1}]$, a LSTM-based RNN network is used to extract the temporal feature h'_{t+s-1}. Finally, a SoftMax classifier is applied to predict the classification probability of k emotion categories of each EEG epoch. The whole working process of our proposed CASC-CNN-LSTM model is described as follows: Input($s \times n$)-Trans($s \times h \times w$)-Conv($s \times h' \times w' \times 32$)-Conv($s \times h'' \times w'' \times 64$)-Conv($s \times h''' \times w''' \times 128$)-FC($l$)-LSTM($s \times i$)-LSTM($s \times i$)-FC($l$)-SoftMax($k$). Here, Input ($s \times n$) denotes raw 1D EEG recordings with s time points and n channels. Trans ($s \times h \times w$) means transforming 1D EEG vector sequence into 2D mesh-like matrix sequence with length of s and size of $h \times w$. Conv ($s \times h' \times w' \times m$) means learning m feature maps size of $h' \times w'$ from each mesh matrix by a convolution layer. FC(l) is a full connection layer with l neurons. LSTM ($s \times i$) indicates an RNN layer with s LSTM cells and the hidden state size of each LSTM cell is i. SoftMax(k) is used to predict k emotion categories.

2.3. Parallel hybrid convolutional recurrent neural network

Fig. 3 shows the architecture of our proposed parallel hybrid convolutional recurrent neural network (Para-CNN-LSTM for short). It is still composed of CNN and RNN. However, unlike the above cascaded model, the RNN here uses each original 1D chain-like EEG epoch as the input, and take the hidden state of the last LSTM cell as the output which will concatenate with the spatial feature representation learned by the parallel running CNN model to get the fused spatiotemporal feature as the input of the final SoftMax classifier to predict the emotion type of each EEG epoch. In the parallel architecture, RNN model is responsible for learning the high-level temporal context information. The jth EEG epoch fed into RNN is represented as: $R_j = [X_t, X_{t+1}, \dots, X_{t+s-1}]$, X_t represents the EEG vector at time point t, and s represents the number of time points in each epoch. The RNN here also contains two LSTM layers, and the number of LSTM cells in each layer is same as that in the cascaded model, because the window

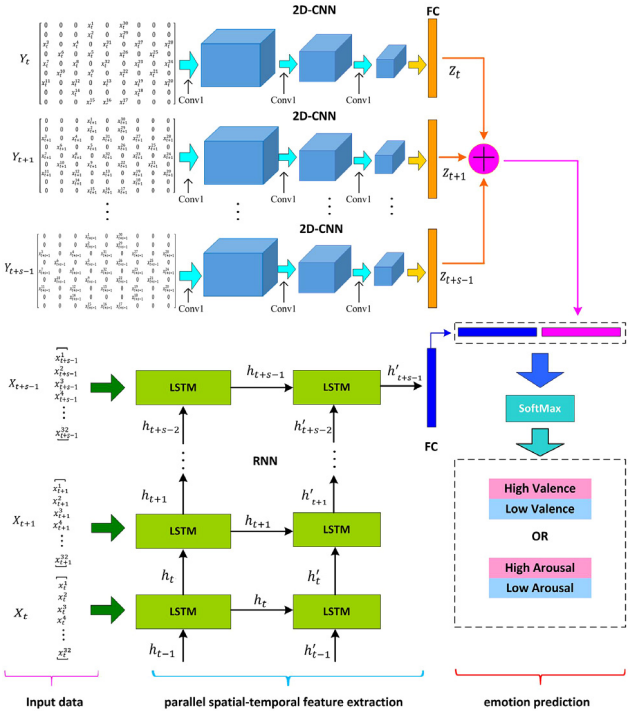


Fig. 3. The architecture of parallel hybrid convolutional recurrent neural network.

size of the input EEG epoch is same in both cases. The hidden state of the last LSTM cell is taken as the output of RNN for further analysis: $h'_{t+s-1} = RNN_{lstm}(R_j)$, $h'_{t+s-1} \in R^i$, where i denotes the size of the hidden state of an LSTM cell. After two LSTM layers, a full connection layer is used to enhance the temporal information representation. The final temporal feature extracted from RNN is represented as: Para-RNN: $O_j = FC\left(h'_{t+s-1}\right)$, $O_j \in R^l$, where l denotes the number of neurons in the last full connection layer of the RNN model.

In Para-CNN-LSTM framework, CNN is still responsible for extracting spatial features, which uses the same input (2D matrix sequence P_j), structure and configuration as those of the CNN in cascaded model. For comparison, we accumulate the spatial feature Z_k extracted from each time point in an epoch to obtain a final single feature vector F_j:Para-CNN: $F_j = \sum_{k=t}^{t+s-1} Z_k$ ($F_j, Z_k \in R^l$), where l is the number of neurons in the last full connection layer of CNN, the same with that in RNN. Then the spatial and temporal features extracted synchronously by CNN and RNN are fused into a joint feature vector, which is fed into the final SoftMax classifier to predict the emotional category: SoftMax: $C_j = SoftMax([F_j, O_j])$, $C_j \in R^k$. According to the previous definition rules, the working process of our Para-CNN-RNN model is described as: ((Input($s \times n$)-Trans($s \times h \times w$) -Conv($s \times h' \times w' \times 32$)-Conv($s \times h'' \times w'' \times 64$)-Conv($s \times h''' \times w''' \times 128$)-FC($l$)-Sum($l$)) ‖ (Input($s \times n$)-LSTM($s \times i$)-LSTM($s \times i$)-FC($l$)))-Fusion($l$)-SoftMax($k$), Where ‖ denotes parallel processing.

In the 2D CNN model of both cascaded and parallel hybrid framework, there is no pooling operation behind each convolution layer. Although in CNN architecture, a convolution layer is usually used in pairs with a pooling layer, it is not necessary. Pooling is to reduce the dimension of data at the expense of losing some information. However, in the EEG data analysis, the data dimension is much smaller than that used in computer vision. In order not to lose useful EEG information, we do not use pooling operation, but directly connects three convolution layers.

3. Experiments and results analysis

We focus on the DEAP dataset to evaluate our proposed cascaded and parallel hybrid convolutional recurrent networks for EEG-based within-subject binary emotion classification. We compare our models against the previous reported methods and pure CNN and LSTM models to show our performance advantages. Additionally, the in?uence of spatial and temporal context information on classification performance is investigated through comparison experiments. Finally, the different variants of both cascaded and parallel models are studied and discussed systematically.

3.1. DEAP dataset

The DEAP dataset a large-scale open-source dataset containing multiple physiological signals with emotional evaluations developed by the research team of Queen Mary University in London for emotion recognition [19]. It recorded the EEG, ECG, EMG and other bioelectrical signals of 32 subjects induced by watching 40 one-minute music videos of different emotional tendencies. The subjects then evaluated the videos' emotion category on scale of 1–9 in dimension of arousal, valence, liking, dominance and familiarity. Valence reflects the degree of people's pleasure, the higher the value, the higher the pleasure degree. Arousal reflects the intensity of people's emotion, the greater the value, more intense and perceptible the emotion. The rating value from small to large indicated the emotion metric was from negative to positive or from weak to strong. The 40 stimulus videos consist of 20 high valence/arousal stimuli and 20 low valence/arousal stimuli. In this paper, we extract the first 32-channel EEG signals of 32 subjects' 40 trails from DEAP dataset to make experiments.

3.2. Data preprocessing

The EEG signals are then pre-processed by down sampling to 128 Hz, band-pass filtering to 4–47 Hz, common average referencing and ocular artifacts removing by blind source separation algorithms. The duration of the denoised EEG signals in each trail is 63-s, including 60-s watching video and 3-s before watching, which have a total of 8064 readings for each channel. We extract 60-s EEG sequence induced by watching video for further analysis. 3 s baseline signals happening before watching videos are removed to correct the changes of stimulus-independent signals and the stimulus-related brain dynamics are acquired. Each sequence is then divided into non-overlapping epochs with a window length of 1 s. As a result, 60 epochs are obtained in each trail and there are total $40 \times 60 = 2400$ EEG epochs in 40 trials of each subject. So, the dataset of each subject is dimension of $2400 \times 128 \times 32$, containing 2400 epochs, 128 time points (also the window size) in each epoch, and 32-channel EEG data at each time point. We call this formatted EEG data as RAW feature. It is found that the performance of both cascaded and parallel hybrid models will be reduced when using window larger or smaller than 128 for sequence segmentation, which may be due to the cycle of human brain emotional activity. Then the data of each raw EEG epoch is normalized across channel to get the NORM features of each subject.

Studies in neuroscience and psychology have shown that EEG signals contain a lot of rhythmic information closely related to psychological activities, such as emotions, in five frequency bands of delta (1 Hz–4 Hz), theta (4 Hz–8 Hz), alpha (8 Hz–13 Hz), beta (13 Hz–30 Hz) and gamma (above 30 Hz) [20]. Our previous research [17] also found that the classification performance of the power spectral density (PSD) feature of EEG data in 1–47 Hz frequency band on 2D CNN network was significantly better than those of raw and norm EEG features in time domain, which may be because the PSD feature extraction in frequency domain needs to be performed in a larger continuous sampling period, the emotional EEG task is also a relatively long-term brain activity and PSD features can probably capture more emotion-correlated brain dynamics. So, on the basis of norm features, we use Fast Fourier algorithm to extract 64 PSD features by sliding 0.5-s Hamming window with no overlap on each channel of 1-s segmented EEG epochs in the whole frequency band of 1–47 Hz. The obtained PSD features in 40 trials of each subject is dimension of $2400 \times 64 \times 32$, which is called FREQ features for the following classification experiments.

Next, based on each subject' affective evaluation of each trial in the range of 1–9 in arousal and valence, the median 5 is used as the threshold to divide the rating value into two categories: more than 5 labeled with 1 meaning high arousal/valence, less than or equal to 5 labeled with 0 meaning low arousal/valence. Thus, total 2400 labels corresponding to EEG epochs is obtained for each subject. The dataset is then balanced to ensure that the number of two types of labels and epochs are equal or close to reduce the impact of data imbalance on classification results, as well as improve the normalization ability of classification models.

3.3. Data transformation

According to the EEG electrode map of DEAP dataset, the 1D chain-like vector sequence of each EEG epoch is converted into 2D mesh-like matrix sequence with the size of 9×9 as formula (1). After data transformation, the EEG epoch contains rich information in space, time and frequency domain. The data and label format of three type of EEG features are shown in Table 1. Then, we randomly extract 75% of the EEG epochs and labels from two categories to train and verify our models, and the rest 25% of the EEG epochs are used to test the training models. In this way, 10-folder cross validation set of PSD features for each subject is constructed. It has been proved by experiments that such partitioning of cross validation set could achieve better classification performance and better generalization.

3.4. Compared models

3.4.1. State-of-the-arts

We will briefly introduce the compared state-of-the-arts models for EEG emotion recognition on the same DEAP dataset.

Alhagry et al. [14] used a deep RNN model based on LSTM structure to classify two types of emotions in arousal, valence and liking dimensions on DEAP dataset. The average binary classification accuracy was 85.65%, 85.45% and 87.99%, respectively.

In our previous study [17], we proposed to apply the shallow Bagging Tree (BT) model to make binary emotion classification on PSD-based EEG features of DEAP dataset in valence and arousal, which accuracy arrived 86.31% and 86.18%, respectively.

Salama [16] et al. suggested a 3D-CNN model for emotion recognition based on multi-channel EEG data of DEAP dataset. The accuracy of binary emotion recognition in valence and arousal was 87.44% and 88.49%, respectively.

We also explored the deep 2D-CNN model [17] to learn the invariable high-level representations from the temporal, frequency and their combined features of EEG signals in DEAP dataset and achieved the binary within-subject recognition accuracy of 88.5% and 86.7% in valence and arousal, respectively.

3.4.2. Baseline models

We also compare our proposed hybrid models with several variants of CNN and RNN models. We apply 1D CNN (without temporal or spatial information), 2D CNN (only with spatial information) and 3D CNN (with spatiotemporal information) for comparing and investigating the influence of spatial and temporal information on brain emotion recognition [12]. The 1D CNN model is fed with raw 32-channel EEG vectors. The 2D CNN model uses the 2D mesh-like matrices transformed by formula (1) as input, without sliding window for segmentation. The input of the 3D CNN model is same as that fed into our proposed

Table 1
The data and label format of three type of EEG features.

Feature style	Feature type	Window size	Feature shape	Label shape
1D chain-like	RAW/NORM	1-s	2400 (epochs) × 128 (points) × 32 (channels) × 1	2400 × 1
	PSD	1-s	2400 (epochs) ×64 (points) × 32 (channels) × 1	2400 × 1
2D mesh-like	RAW/NORM	1-s	2400 (epochs) × 128 (points) × 9 (width) × 9 (height)	2400 × 1
	PSD	1-s	2400 (epochs) ×64 (points) × 9 (width) × 9 (height)	2400 × 1

cascaded hybrid model. These three CNN models all consist of three convolution layers without pooling, a full connection layer with 1024 neurons and a SoftMax output layer. The convolutional kernel size of 1D, 2D and 3D CNN models are 3, 3 × 3 and 3 × 3 × 3, respectively. The step size is 1 without padding. The number of feature maps of each convolution layer is 32, 64 and 128 for all three baseline CNN models. For fare comparison, we set all the hyperparameters of three baseline CNN models to be consistent with those of CNN in our proposed models, except for special ones of the hybrid models themselves.

The neural networks studied in this paper are all implemented on TensorFlow framework and Nvidia Titan X Pascal GPU in a fully supervised way from scratch. The cross-entropy loss function of the model is minimized by Adam based random gradient descent algorithm. 1024 neurons are used in all fully connected layers. To ensure the fairness of the comparison, the baseline RNN models also have two LSTM layers before a fully connection layer and the hidden state size of LSTM cell is set 64 and 16, respectively.

3.5. Experimental results

In this part, we will show the overall performance of our proposed models and the comparison results and systematically analyze the influence of EEG spatial and temporal context information on the experimental results.

The overall experimental performance of our proposed Casc-CNN-LSTM model, Para-CNN-LSTM model and several compared baseline models are presented in Table 2. It is found that both our cascaded and parallel models have achieved higher accuracy of more than 93%, which are superior to the state-of-the-arts methods and baseline models. Generally speaking, the classification accuracy of Casc-CNN-LSTM model either in valence or in arousal is slightly higher than that of Para-CNN-LSTM model, but the difference is not significant. For both cascaded and parallel hybrid models, their classification accuracies in valence are higher than those in arousal. Moreover, our 3D CNN baseline model achieves better performance than the state-of-the-arts methods. This indicates that the context spatial and temporal information is very important to boost the performance of EEG-based emotion classification and recognition. In our experiments, we also try to use a larger or smaller sliding window to segment the EEG data, and the performance of two hybrid models in both cases decrease significantly. our models have better flexibility to adapt to different types of EEG-base emotion recognition by changing the size of sliding window over different range. Moreover, compared with the previous methods, our models need less preprocessing on the raw EEG data, which is more suitable for real-time BCI applications.

3.6. Influence of spatial and temporal context information

To study the impact of spatial and temporal information on EEG-based emotion recognition and compare our proposed hybrid models with other baseline neural networks, we construct three alternative CNN models as described above, and their classification performance is shown in Table 2. While comparing 1D CNN with 2D CNN, it is found that 2D CNN outperforms 1D CNN, which implies that extracting EEG features from 2D mesh-like physical neighboring sensors can capture more spatial correlated information than from simple 1D chain-like EEG vectors and it is more helpful to recognize the emotional patterns of human brain. Additionally, the emotional dynamics of human brain can also be defined as a series of periodic activities. 3D CNN has been widely used to extract local spatiotemporal information of sequential signals [21]. Therefore, we also use 3D CNN to learn the spatial and temporal high-level discriminative information from EEG mesh sequences synchronously. It can be seen from Table 2 that the classification accuracy of 3D CNN is respectively 11.1% and 11.3% higher than that of 1D CNN and 7.2% and 9.1% higher than that of 2D CNN in valence and arousal, showing a consistent improvement compared with 1D CNN and 2D CNN.

While our hybrid Casc-CNN-LSTM model uses LSTM rather than CNN to further learn the global temporal dynamics between consecutive sampling points in an EEG epoch after extracting the spatial feature representations from 2D meshes of each time point. As expected, its classification accuracy is further improved, which is 2.2% and 1.7% higher than that of 3D CNN in valence and arousal, respectively. However, when the pure LSTM model which just considers the term correlation is used, the recognition accuracy is reduced to about 80%, which indicates again that the jointly learned temporal and spatial features are crucial for EEG-based emotion recognition.

We observe the similar results when testing our Para-CNN-LSTM model. It outperforms both 2D CNN which just considers extracting spatial context information and LSTM (64 or 16) model which just considers extracting temporal context information. Its classification accuracy in valence and arousal is respectively 8.9% and 10.7% higher than that of 2D CNN, and 11.4% and 12.1% higher than that of LSTM (64) model. In summary, the recognition performance of both cascaded and parallel hybrid models is over 93%, indicating that our proposed two joint feature extraction methods have strong spatiotemporal context representation ability and the learned spatiotemporal fusion features have significantly enhanced the performance of EEG-based emotion recognition.

3.7. Structure analysis of cascaded hybrid models

There are many cascaded model variants. Since it is impossible to exhaustively study all possible structures in detail, we only investigate the key components of the cascaded model. In the experiment, it is found that when the hidden state size of LSTM cell is 64, the cascaded structure produces the best performance. This may be because it is difficult to train the networks with larger hidden states due to the limitations of experimental conditions and the smaller hidden state have limited ability to represent features. Therefore, the following variants of the cascaded model are all based on LSTM cells with hidden state size of 64. The performance comparison of these cascaded model variants is summarized in Table 3. The results show that the more layers CNN or RNN has, the higher classification accuracy acquired. However, this performance improvement is at the cost of computing resources, so we decide to use three CNN layers and two LSTM layers to make balance between these two factors. Full connection layer especially the one connecting CNN and LSTM is very important for overall performance, which indicates that before input into another type of neural network, reconstruction of the previous network output is helpful to the following feature extraction. However, the contribution of the final fully connected layer is not significant, which probably due to the fact that two LSTM layers have learned spatiotemporal features robust enough to predict the final emotion category.

Table 2
Comparison of binary emotion classification accuracy in valence and arousal between the baseline models and our proposed models. LSTM (64) and LSTM (16) indicate that the size of hidden state of LSTM cells of two baseline RNN models is 64 and 16, respectively.

Method	Input features	Valence Acc	Arousal Acc
Alhagr [14] LSTM	Segmented 1D Raw chain sequence	0.8545	0.8565
Chen [17] BT	Segmented 1D PSD chain sequence	0.8631	0.8618
Salama [16] 3D-CNN	Segmented 1D Raw chain sequence	0.8744	0.8849
Chen [17] 2D-CNN	Segmented Norm+PSD chain sequence	0.8853	0.8666
1D-CNN	1D PSD chain sequence	0.8036	0.8022
2D-CNN	2D PSD mesh sequence	0.8429	0.8238
3D-CNN	Segmented 2D PSD mesh sequence	0.9149	0.9152
LSTM(64)	Segmented 1D PSD Chain sequence	0.8176	0.8094
LSTM(16)	Segmented 1D PSD Chain sequence	0.7631	0.7695
Casc-CNN-LSTM	**Segmented 2D PSD mesh sequence**	**0.9364**	**0.9326**
Para-CNN-LSTM	Segmented 2D PSD mesh sequence, Segmented 1D PSD Chain sequence	0.9317	0.9305

Table 3
Performance comparison of cascaded model variants.

Cascaded structure	Acc on valence	Acc on arousal
1-layer CNN+FC+2-layer LSTM+FC	0.8812	0.8820
2-layer CNN+FC+2-layer LSTM+FC	0.9138	0.9196
3-layer CNN+FC+2-layer LSTM+FC	**0.9364**	**0.9326**
3-layer CNN+2-layer LSTM+FC	0.8757	0.8731
3-layer CNN+ FC+2-layer LSTM	0.9249	0.9243
3-layer CNN+ FC+1-layer LSTM+FC	0.9336	0.9310

Table 4
Performance comparison of parallel model variants.

Parallel structure	Fusion method	Acc on valence	Acc on arousal
1-layer CNN+FC 2-layer LSTM+FC	Concatenate	0.8759	0.8801
2-layer CNN+FC 2-layer LSTM+FC	Concatenate	0.9214	0.9233
3-layer CNN+FC **2-layer LSTM+FC**	Concatenate	**0.9317**	**0.9305**
3-layer CNN+FC 1-layer LSTM+FC	Concatenate	0.9283	0.9269
3-layer CNN+FC 2-layer LSTM+FC	Add	0.9312	0.9304
3-layer CNN+FC 2-layer LSTM+FC	Cat+FC	0.9150	0.9158
3-layer CNN+FC 2-layer LSTM+FC	Cat+Conv	0.9125	0.9132

3.8. Structure analysis of parallel hybrid models

Table 4 shows the performance comparison result of different parallel model variants. The hidden state size of LSTM cells in RNN part is set 64 according to previous experience. The performance improvement is also observed with the increase of CNN layers and LSTM layers. The most significant difference between the cascaded model and the parallel model is in the fusion method of spatial and temporal features. In the parallel model, EEG data is learned by CNN and LSTM model at the same time, so there are many different approaches can be used to fuse the parallel learned features. In this paper, we mainly investigate two basic fusion methods including concatenation and addition and two improved fusion methods such as concatenation joint full connection and concatenation joint point-wise convolution [18]. Interestingly, the basic fusion methods achieve better performance with classification accuracy higher than 93%. Since hybrid neural networks need more subtle training and parameter fine tuning to get better performance, it is not necessary to apply more complex fusion operations when the basic method can get satisfactory results.

4. Conclusions

In this paper, a new spatiotemporal EEG representation method is proposed, which transforms the raw 1D chain-like EEG vector at each time point into 2D mesh-like spatial matrix corresponding to the EEG electrode map and thus better representing the spatial correlation among multiple physically neighboring EEG electrodes. Then, the 2D meshes sequence is segmented into epochs of equal length by sliding window method, which produces new EEG representations integrating the temporal and spatial correlated information. Two trainable cascaded and parallel hybrid deep learning models for EEG emotion recognition are proposed. Both of them aim to end-to-end learn the spatial correlated context information through CNN and capture the temporal dependence of EEG data streams through LSTM network. We make within-subject binary emotion classification experiments on large-scale data set DEAP in valence and arousal to evaluate our proposed methods. We also compare the state-of-the-arts and baseline models with our proposed models. The experimental results demonstrate that the recognition accuracies of our two hybrid models are more than 93% both in valence and arousal, which obviously outperform the state-of-the-arts methods. It is proved that our models can jointly learn the spatial and temporal correlated EEG dynamics, which further improve the accuracy and robustness of EEG-based emotion recognition.

The proposed models can be used to assessing emotional responses induced by watching video, listening music or experiencing virtual reality (VR) wearing a consumer multi-electrode EEG headset. The prediction can be implemented by a wearable EMOTIV EPOC+ 14-channel EEG headset and a custom-developed BCI application software. The software wirelessly receives EEG signals from the headset and dynamically stores them in a relational database for subsequent analysis and emotion classification. The recognized emotion state can be sent to another application which may execute different commands according to different emotion states, such as a video/music/VR therapy app in which the player receives user's emotion state recognized by our proposed models and recommends proper video/music/VR to the user according to user's emotion feedback. For example, if the music player receives a sad emotion state from the user, it will choose the happy music to play to cheer up the user. So, our proposed methods and models are valuable for developing affective BCI applications via wearable multi-electrode EEG headset.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jingxia Chen: Writing - original draft, Methodology, Funding acquisition, Writing - review & editing. **Dongmei Jiang:** Investigation, Formal analysis, Software. **Yanning Zhang:** Conceptualization, Supervision, Resources. **Pengwei Zhang:** Data curation, Resources, Visualization.

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